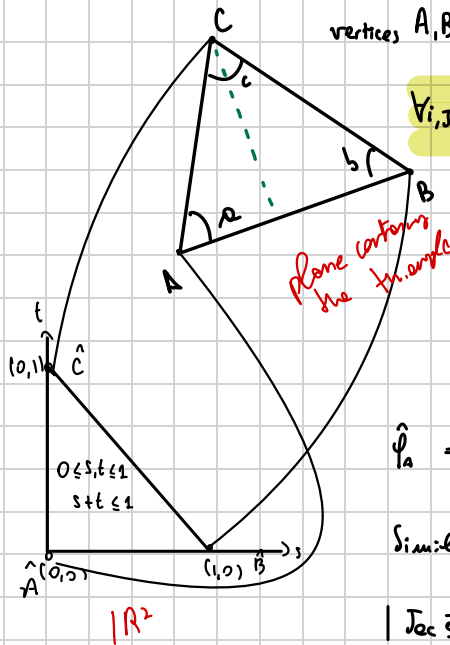


vertices $A, B, C \in \mathbb{R}^3$

change of variable

$$\forall i, j \in \{A, B, C\} \text{ RHN: } \int_{ABC} \varphi_i \varphi_j = \int_{\hat{ABC}} \hat{\varphi}_i \hat{\varphi}_j |\text{Jacobian } \Phi| ds dt$$

$\hookrightarrow \in \mathbb{R}^2$



$$\hat{\varphi}_i(s, t) = A_i + B_i s + C_i t$$

$$\hat{\varphi}_a(s, t) = 1 \quad \text{if } (s, t) = (s_a, t_a) \quad \text{ELSE} = 0$$

$$\hat{\varphi}_b(s, t) = 1 \quad \text{if } (s, t) = (s_b, t_b) \quad \text{ELSE} = 0$$

$$\hat{\varphi}_c(s, t) = 1 \quad \text{if } (s, t) = (s_c, t_c) \quad \text{ELSE} = 0$$

$$\hat{\varphi}_a \Rightarrow \begin{cases} a_a + b_a \cdot s_a + c_a \cdot t_a = 1 \\ a_a + b_a \cdot s_b + c_a \cdot t_b = 0 \\ a_a + b_a \cdot s_c + c_a \cdot t_c = 0 \end{cases} \Rightarrow \hat{\varphi}_a(s, t) = 1 - s - t$$

Similarly solve the linear system to find $\hat{\varphi}_b(s, t) = s$, $\hat{\varphi}_c(s, t) = t$

$$|\text{Jac } \Phi| = \frac{\text{Area}(ABC)}{\text{Area}(\hat{ABC})} = \frac{\text{Area}(ABC)}{(1/2)} = 2 \text{Area}(ABC)$$

$$M_{ij} = \begin{cases} \frac{1}{24} \cdot 2 \text{Area}(ABC) & \text{if } i \neq j \\ \frac{1}{12} \cdot 2 \text{Area}(ABC) & \text{if } i = j \end{cases}$$

"mass matrix"